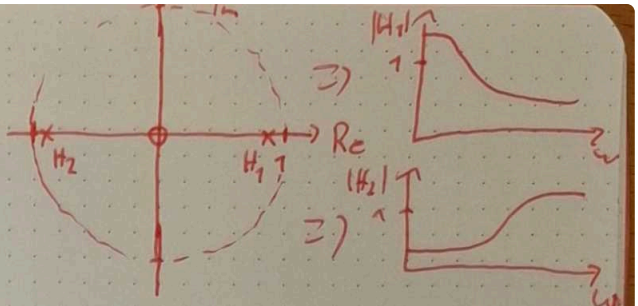
Problem 1 & 2)

1) a) $H(z) = \frac{1}{1 \pm 0{,}9z^{-1}} = \frac{z}{z \pm 0{,}9}$ $\Rightarrow$



We get H_1 = Lowpass and H_2 = Highpass.

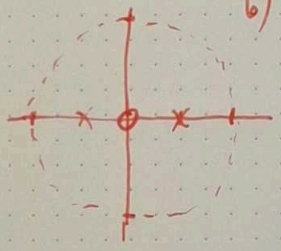
When travelling around the unit circle, both H have a constant distance to the zero in 0, while the distance $|z - z_n|$ changes. When closest to the pole, that frequency gets amplified, while the opposite for when closest to the zero.

b) Checks out with pezdemo from lectures/Fallstad.

2) a) $H(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 + \frac{1}{2}z^{-1})}$ $\Rightarrow 1 = A(1 + \frac{1}{2}z^{-1}) + B(1 - \frac{1}{2}z^{-1})$

$\Rightarrow A = \frac{1}{2} \quad B = \frac{1}{2}$

$\Rightarrow H(z) = \frac{1}{2}\left(\frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 + \frac{1}{2}z^{-1}}\right) = \frac{1}{2}\left(\frac{z}{z - \frac{1}{2}} + \frac{z}{z + \frac{1}{2}}\right)$



b) Since zeros and poles are within unit circle, there is a stable and causal inverse given by

$$H_I(z) = \frac{1}{H(z)} = \underline{\underline{(1 - \tfrac{1}{2}z^{-1})(1 + \tfrac{1}{2}z^{-1})}}$$

c) Yes its a minimum phase filter.

d) No it does not as $H_I(z) = 1 - \frac{1}{4}z^{-2}$

$\Rightarrow h_I[n] = \mathcal{Z}^{-1}\{1\} - \frac{1}{4}\mathcal{Z}^{-1}\{z^{-2}\} = \underline{\delta[n] - \frac{1}{4}\delta[n-2]}$

$\Rightarrow$ We can see that $h_I[n]$ is not symmetric as

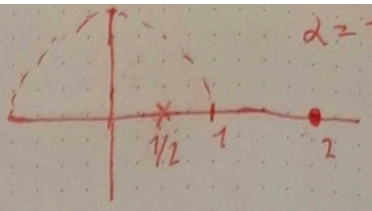
$h_I[1] = 0$ and $h_I[0] = 1 \neq h_I[2] = -\frac{1}{4}$

which is required for a FIR system to have a linear phase charecteristic.

Problem 3)

3) a)

$$A(z) = \frac{\alpha z - 1}{z - \alpha} \Rightarrow \begin{matrix} p_k = \alpha \\ z_k = 1/\alpha \end{matrix}$$



If you do $|A(e^{j\omega})|$ you get 1, so it's a allpass meaning it has constant amplitude response, but varying phase resp.

b) Lower: $\left(x[n] - x[n] * a[n]\right) \cdot \frac{1}{2} \overset{z}{\leftrightarrow} \left(X(z) - X(z)A(z)\right)\frac{1}{2}$

Upper: $\left(x[n] + x[n] * a[n]\right) \cdot 1/2 \overset{z}{\leftrightarrow} \left(X(z) + X(z)A(z)\right)1/2$

$y[n]$ = upper + lower

$$\Rightarrow H_{up}(z) = 1/2\left(1 + A(z)\right) = \frac{1}{2} + \frac{\alpha - z^{-1}}{2(1 - \alpha z^{-1})} = \frac{1 - \alpha z^{-1} + \alpha - z^{-1}}{2(1 - \alpha z^{-1})}$$

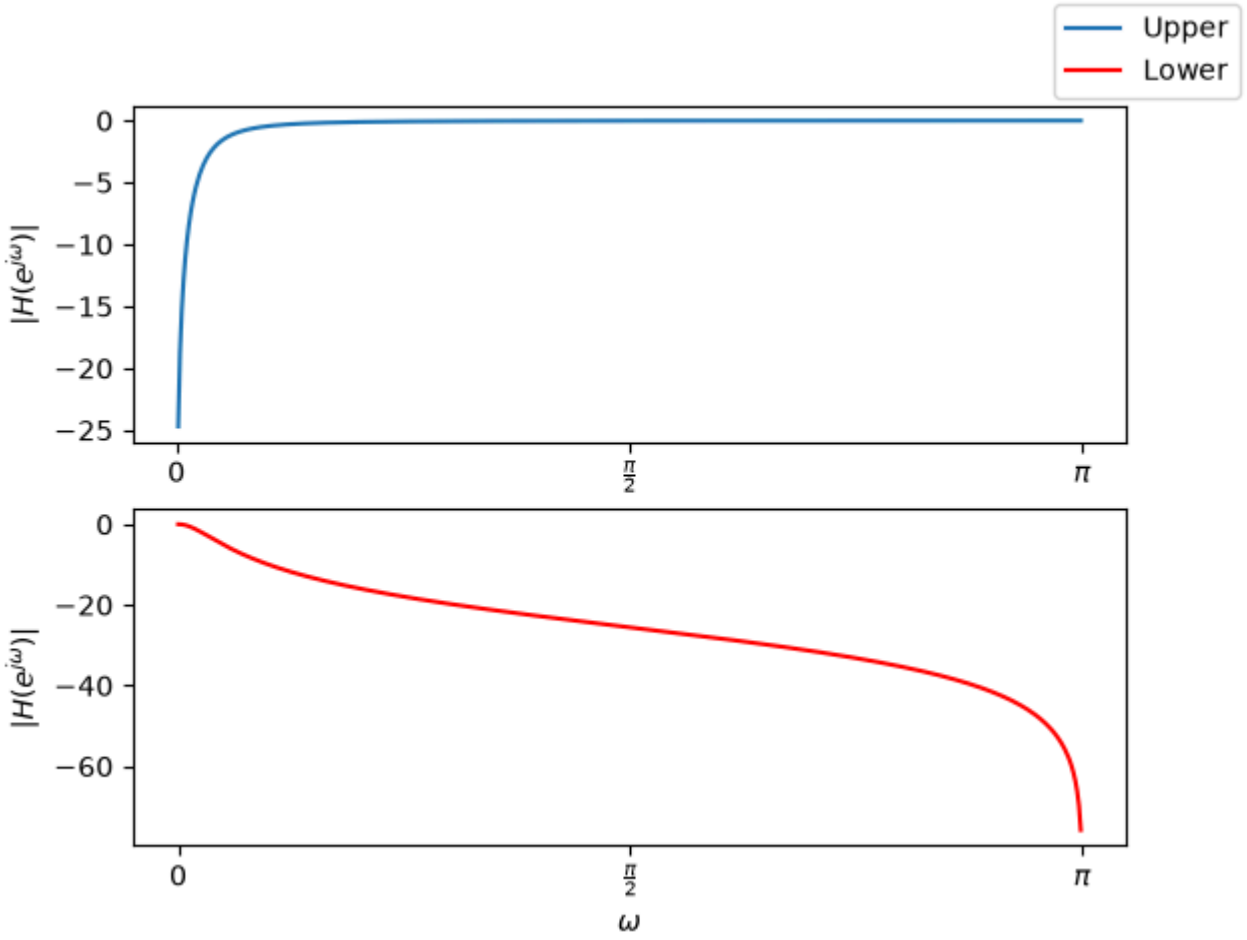
$$\Rightarrow \frac{1 + \alpha - (1+\alpha)z^{-1}}{2 - 2\alpha z^{-1}}$$

$$H_{lo}(z) = 1/2\left(1 - A(z)\right) = 1/2 - \frac{\alpha - z^{-1}}{2(1 - \alpha z^{-1})}$$

$$\Rightarrow \frac{1 - \alpha z^{-1} - \alpha + z^{-1}}{2(1 - \alpha z^{-1})} = \frac{1 - \alpha + (1 - \alpha)z^{-1}}{2 - 2\alpha z^{-1}}$$

b/a

Plotting in python i get clearly a highpass and lowpass filter for upper and lower.



Code:

PYTHON

```python
import matplotlib
matplotlib.use("QtAgg")

from scipy.signal import freqz
import matplotlib.pyplot as plt
import numpy as np

alp = 0.9
K = 1

b_upper = [1+alp, -(1+alp)]
a_upper = [2, -2*alp]

b_lower = [1-alp, 1-alp]
a_lower = [2, -2*alp]

w_upper, h_upper = freqz(b_upper, a_upper)
w_lower, h_lower = freqz(b_lower, a_lower)

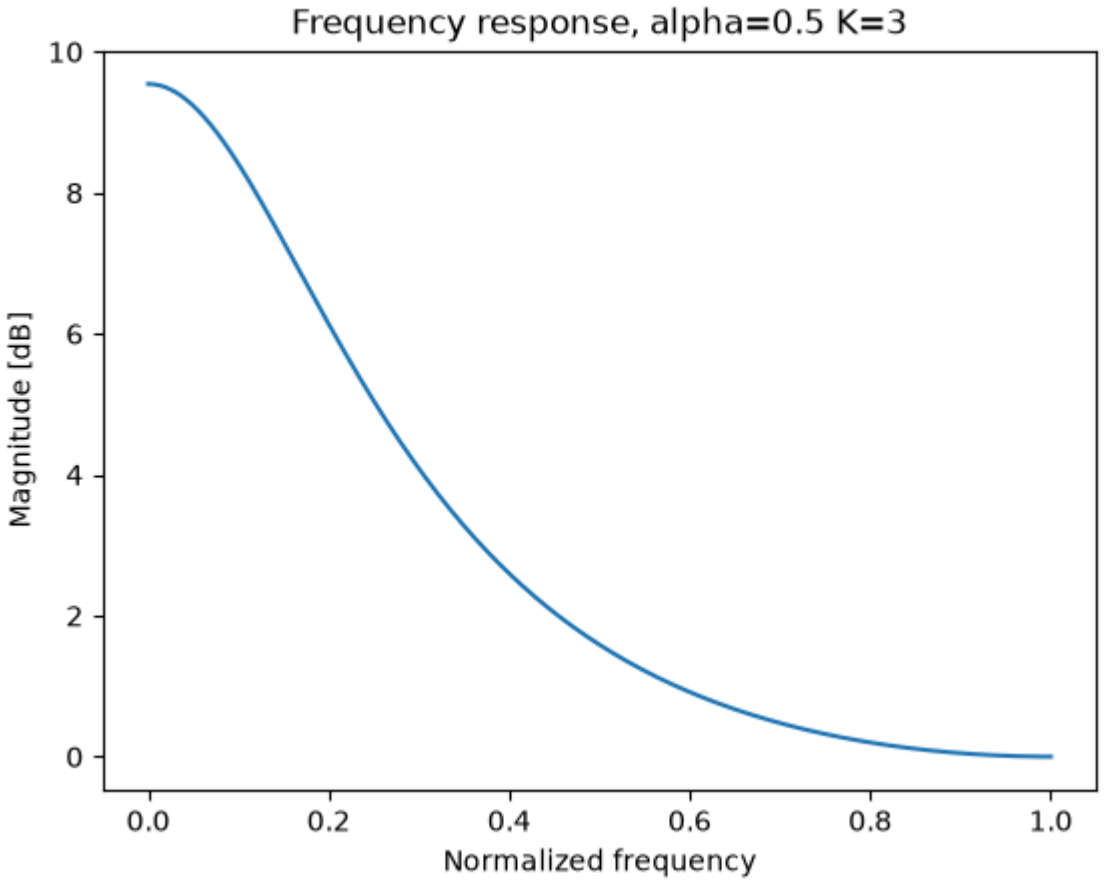
fig, (ax1, ax2) = plt.subplots(2,1,)

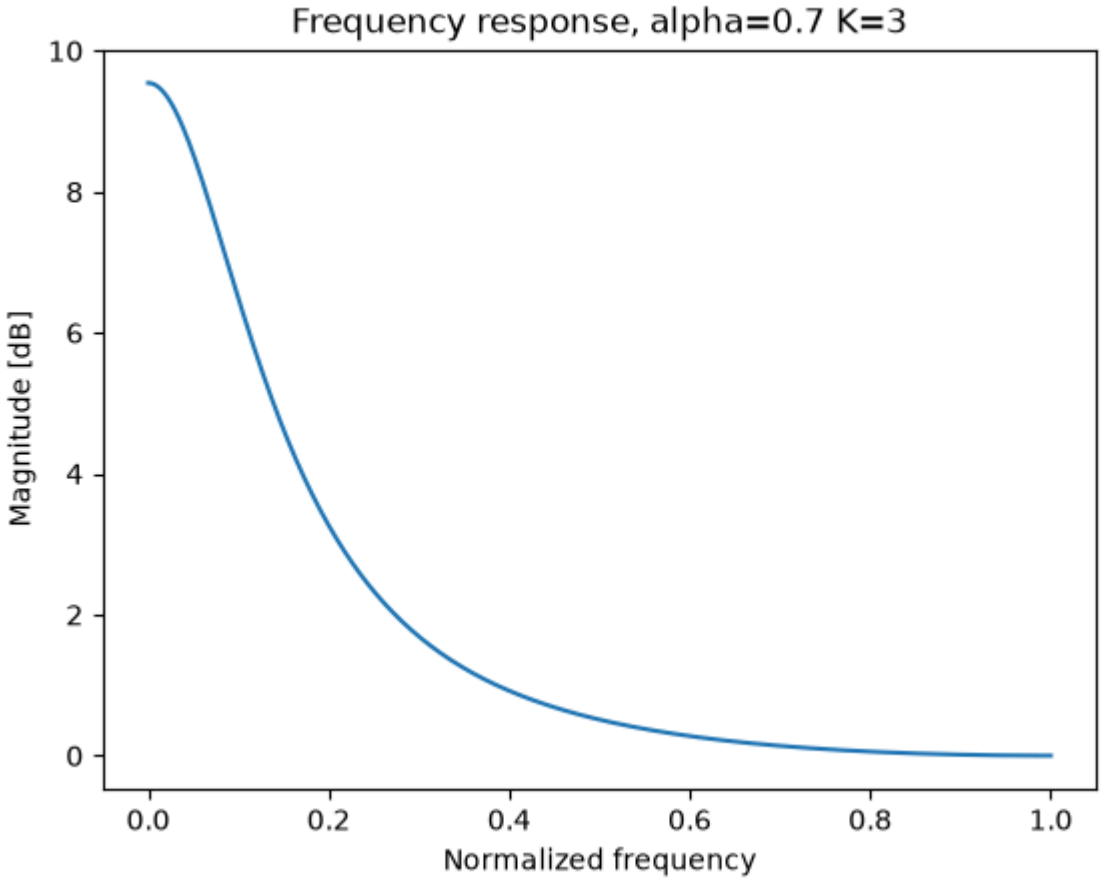
ax1.plot(w_upper, 20 * np.log10(abs(h_upper)), 'C0', label=r"Upper")
#ax1.set_xlabel(r"$\omega$")
ax1.set_ylabel(r"$|H(e^{j\omega})|$")
ax2.plot(w_lower, 20 * np.log10(abs(h_lower)), 'r', label=r"Lower")
ax2.set_xlabel(r"$\omega$")
ax2.set_ylabel(r"$|H(e^{j\omega})|$")

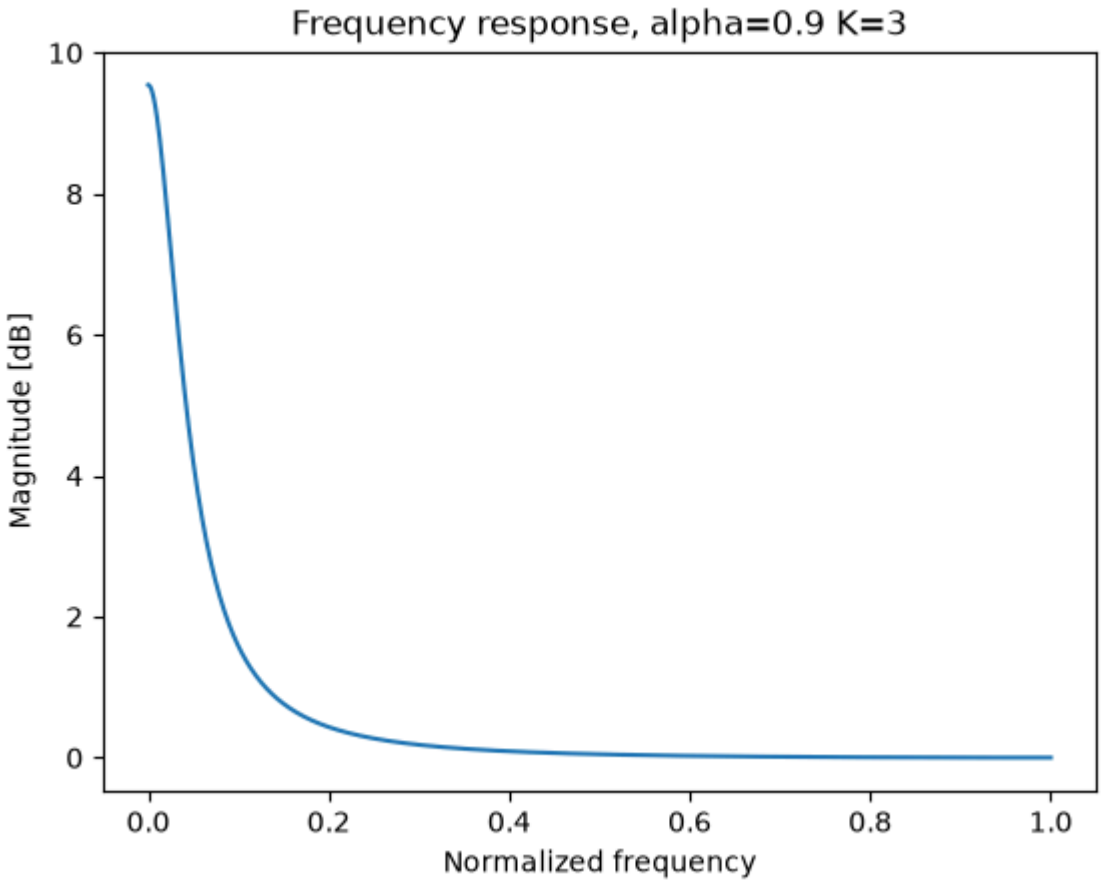
tick_positions = [0, np.pi/2, np.pi]
tick_labels = [r'$0$', r'$\frac{\pi}{2}$', r'$\pi$']
ax1.set_xticks(tick_positions)
ax1.set_xticklabels(tick_labels)
ax2.set_xticks(tick_positions)
ax2.set_xticklabels(tick_labels)

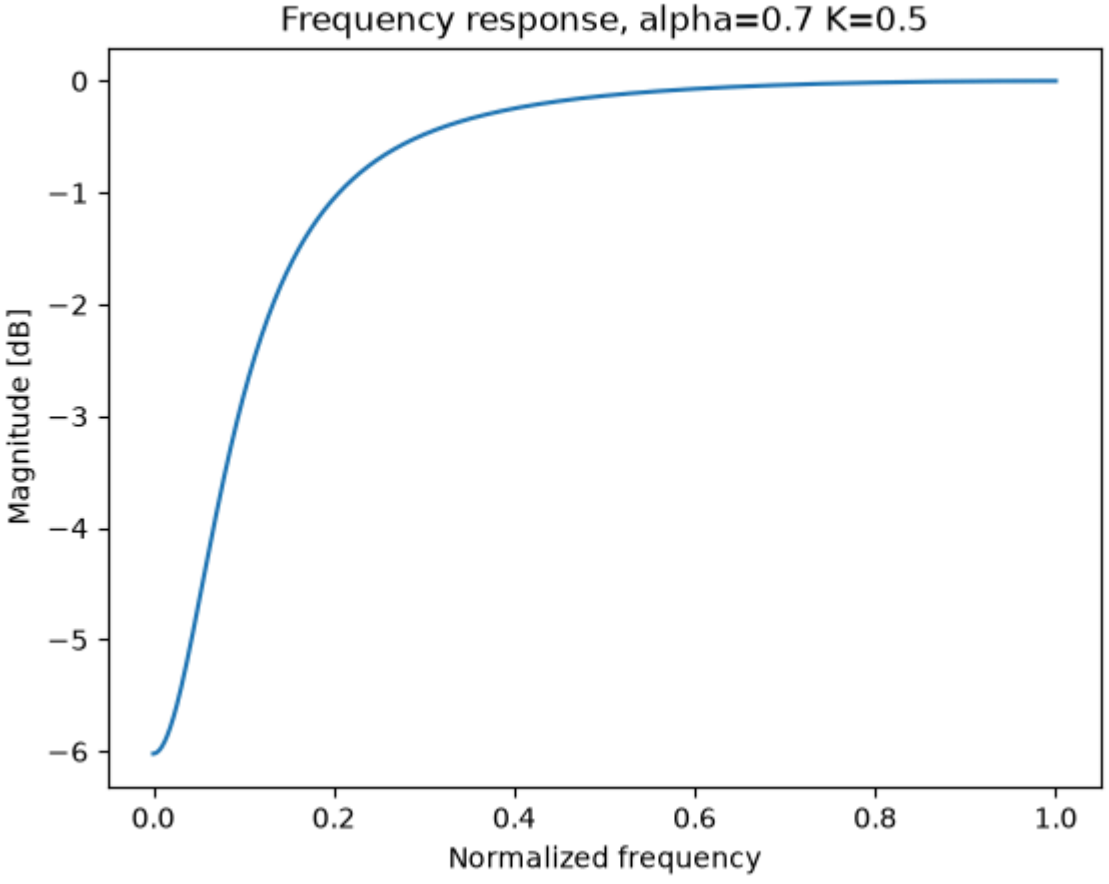
fig.legend()
plt.savefig("task_3_b")
plt.show()
```

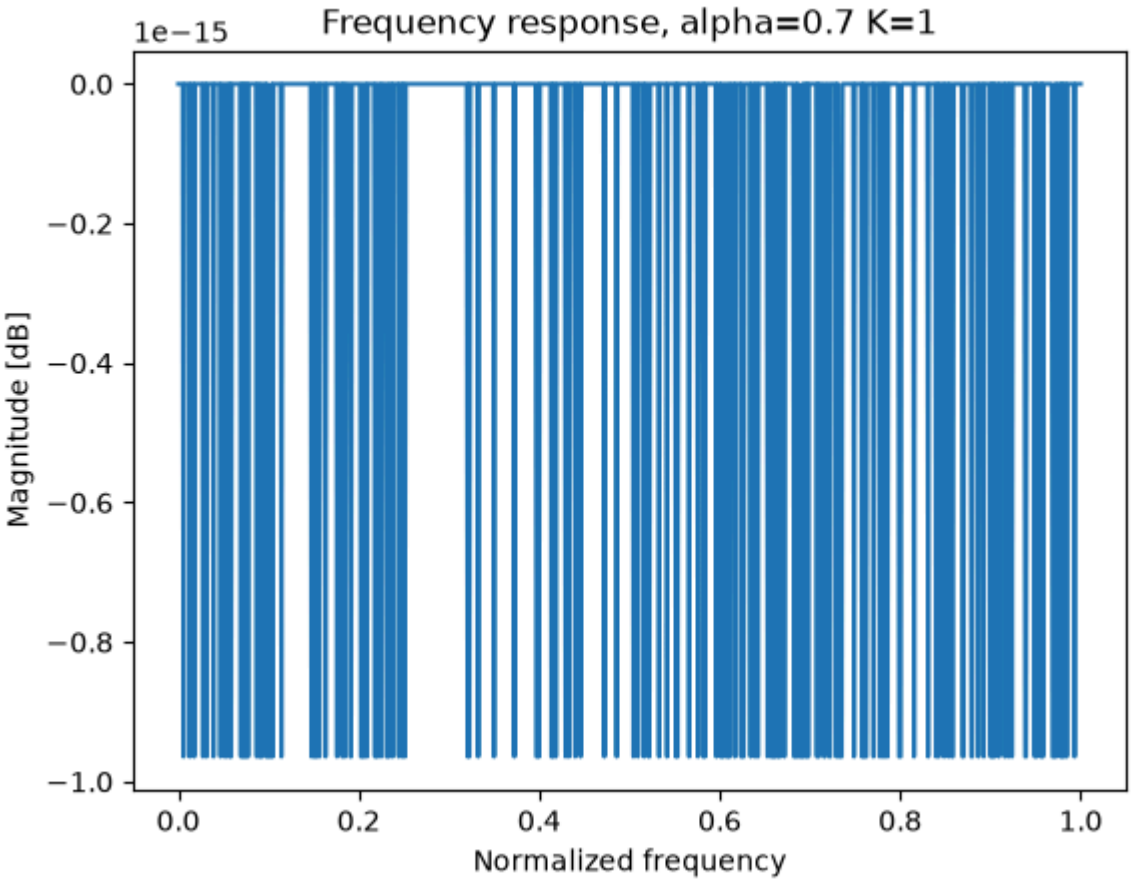
c)

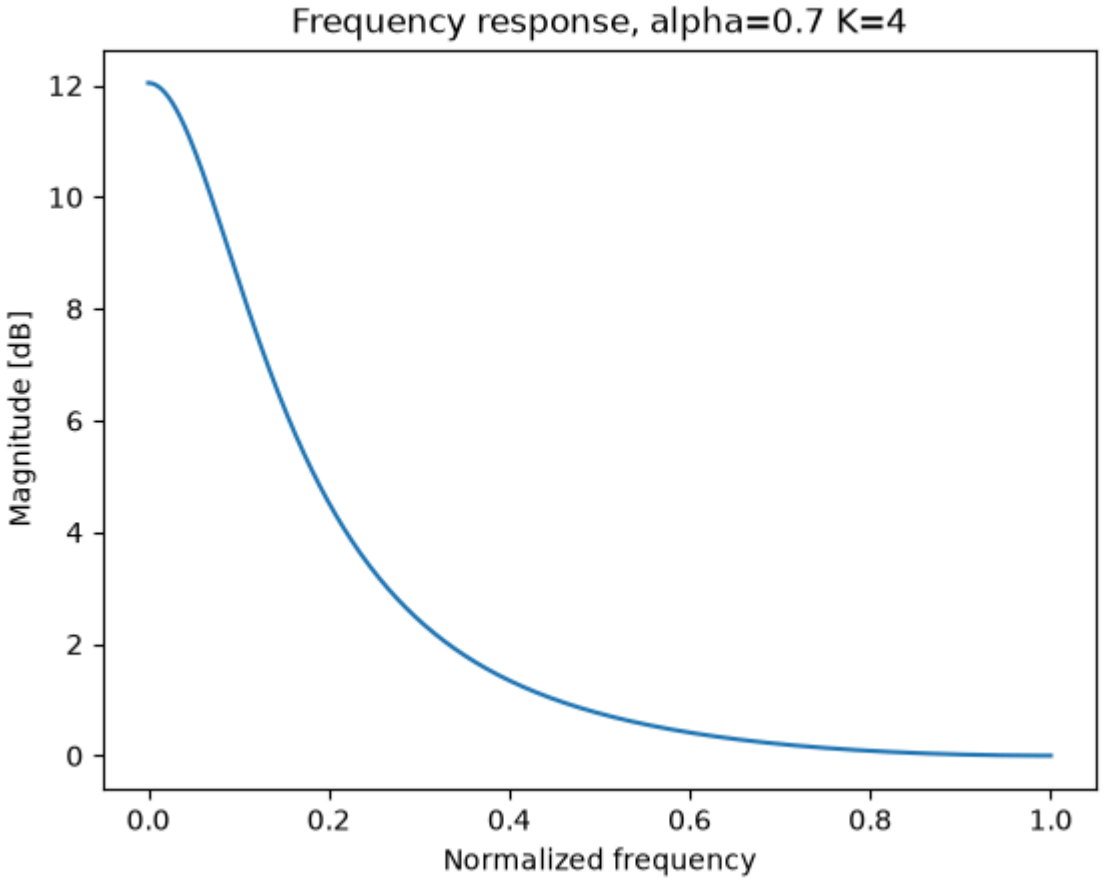

Frequency response, alpha=0.5 K=3
Magnitude [dB]
Normalized frequency
0
2
4
6
8
10
0.0
0.2
0.4
0.6
0.8
1.0


Frequency response, alpha=0.7 K=3
Magnitude [dB]
Normalized frequency
0
2
4
6
8
10
0.0
0.2
0.4
0.6
0.8
1.0


Frequency response, alpha=0.9 K=3
Magnitude [dB]
Normalized frequency
10
8
6
4
2
0
0.0
0.2
0.4
0.6
0.8
1.0


Frequency response, alpha=0.7 K=0.5
Magnitude [dB]
Normalized frequency
0
−1
−2
−3
−4
−5
−6
0.0
0.2
0.4
0.6
0.8
1.0



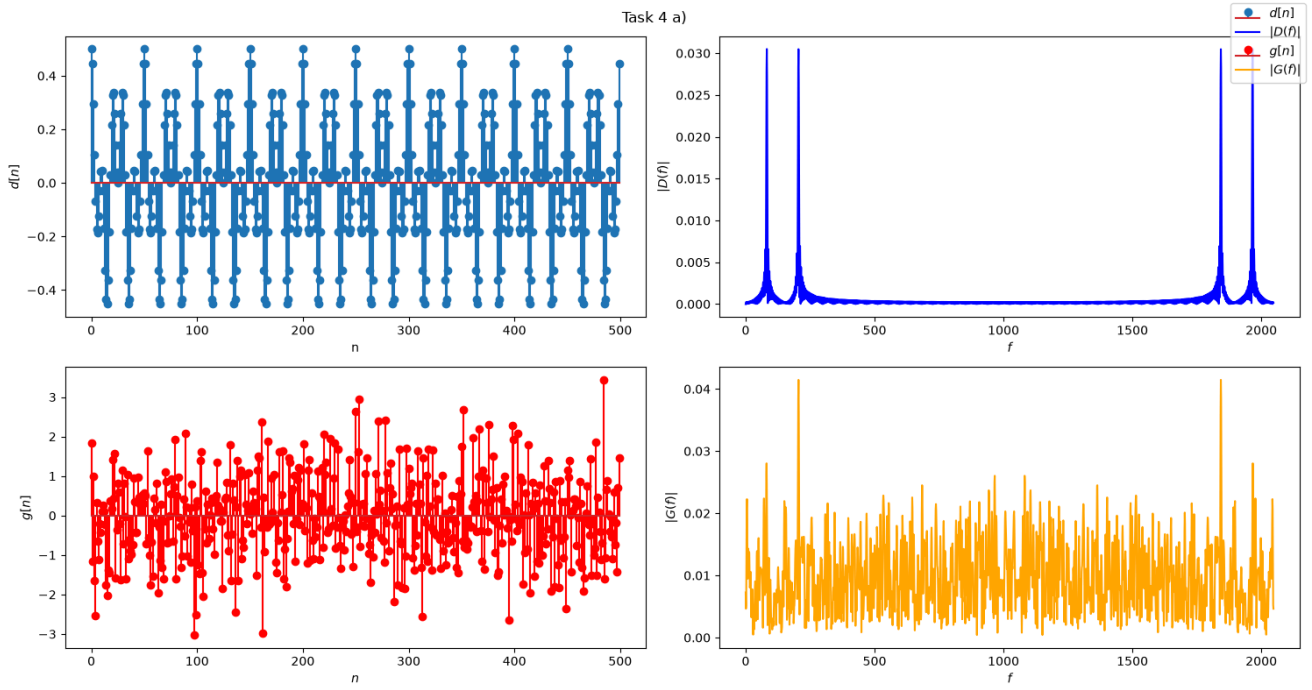


K and α respectively controll the amount of suppression/amplification of the magnitude and how much this changes with frequency (how steep the slope is).

Now please no more listening exercises, I can barely tell the difference and its a pain to setup with python.

Problem 4)

a)



The plots with noice is completely unrecognisable, while the original has two distinct frequencies with its aliasing due to the sampling at 2048.

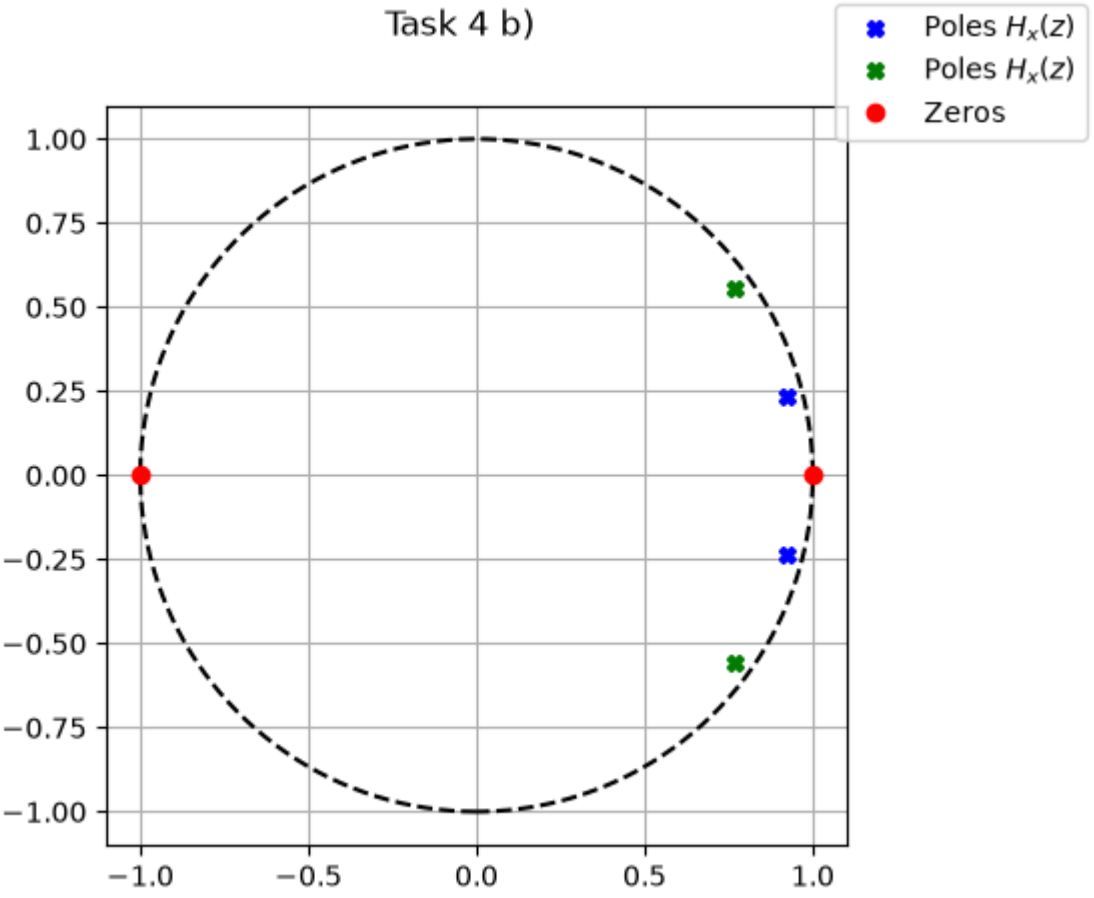
b)

The filters can be written as:

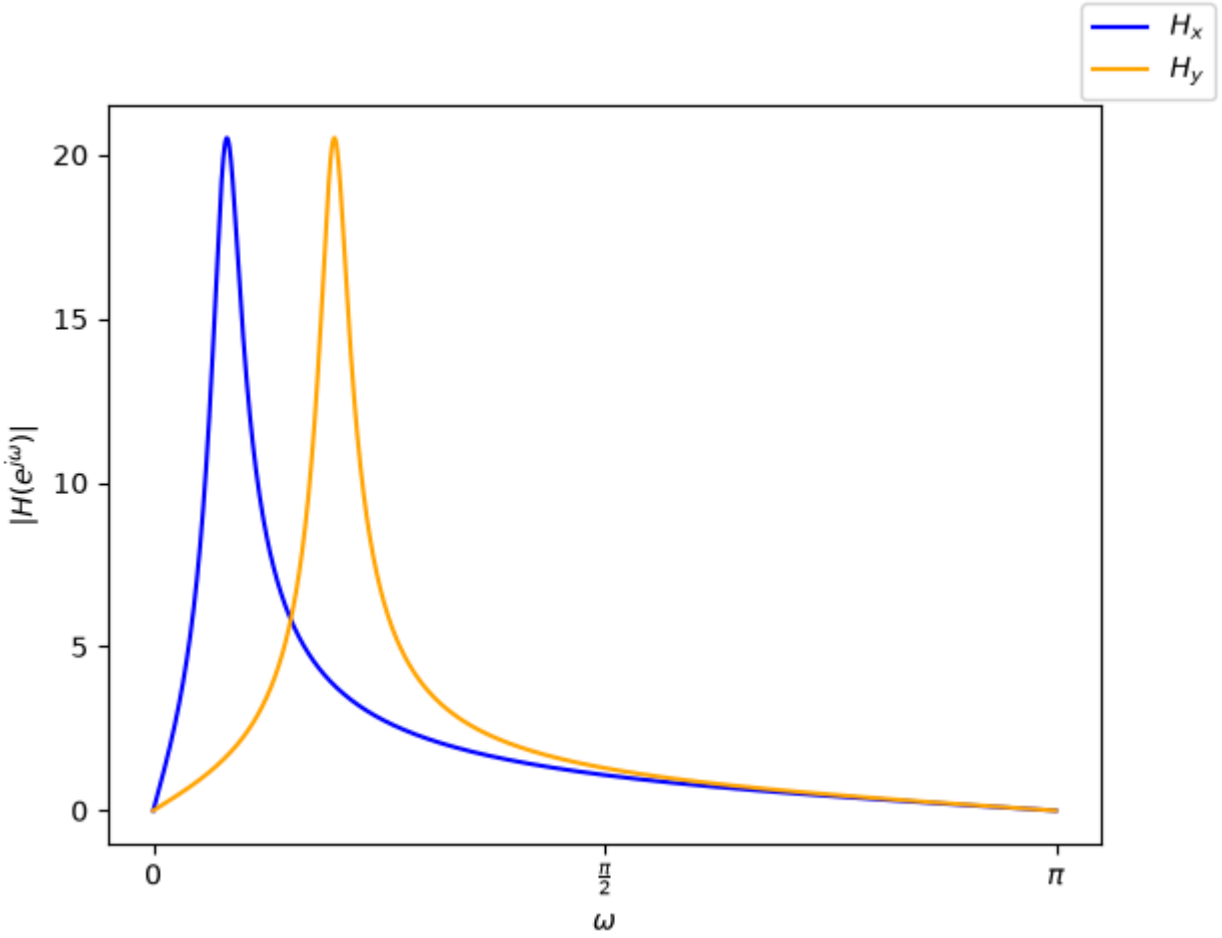
$$H_i(z) = \frac{1 - z^{-2}}{1 - 2r\cos(\omega_{0_i})z^{-1} + r^2 z^{-2}}$$

Where $r = 0.95$, $\omega_{0_i} = f_i \cdot 2\pi$ and $f_1 = 0.04$ and $f_2 = 0.10$.

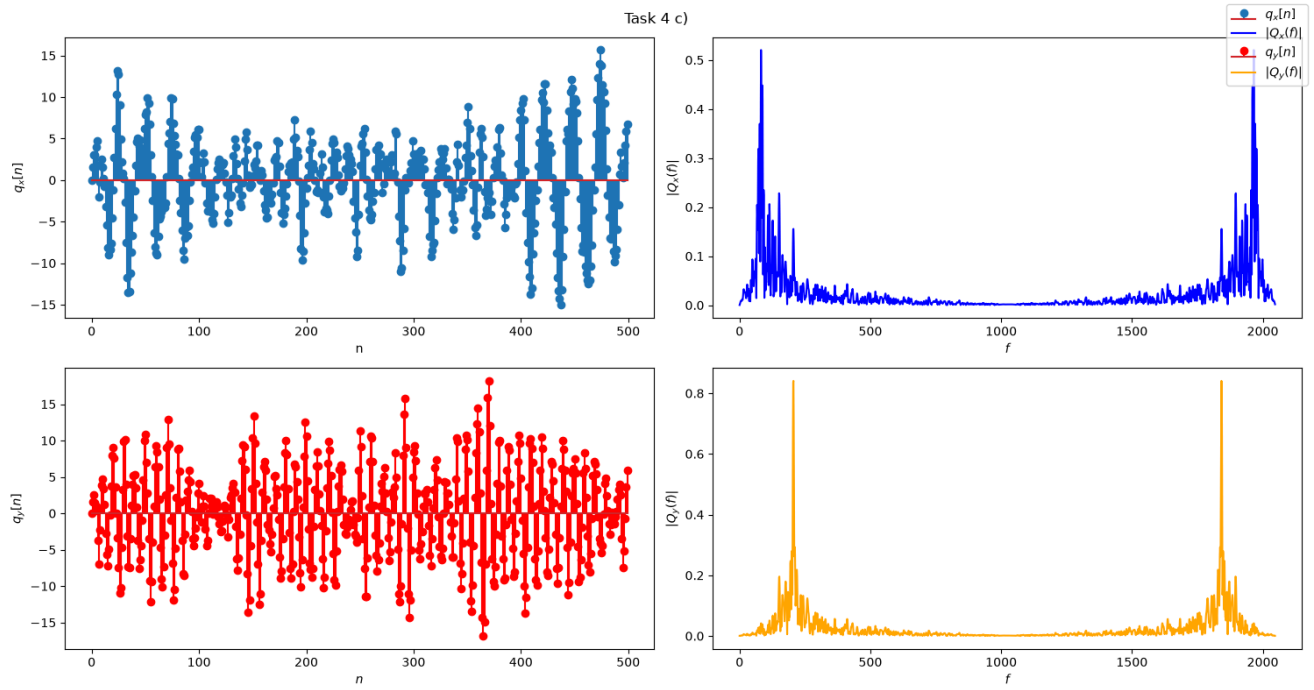
The pole-zero-plot:



Filters H_x and H_y

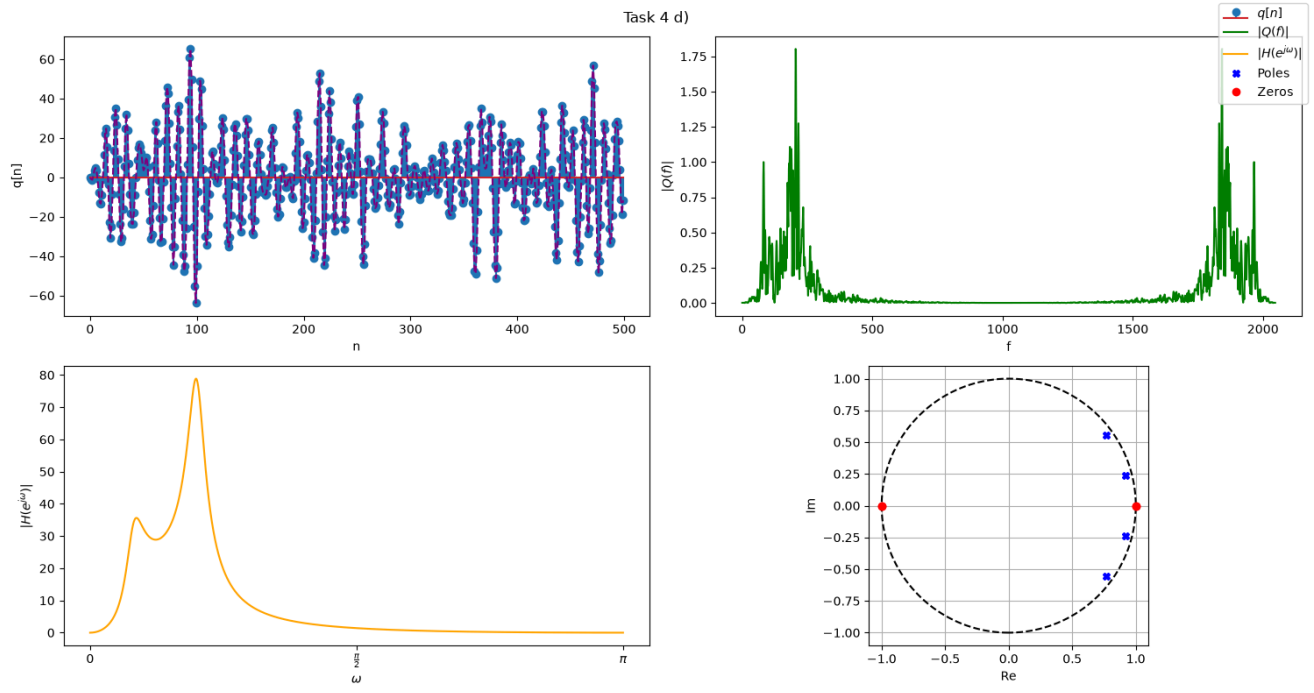


c)



Looks a bit better but still really noisy.

d)



Did not get it to be as good as the original, got alot of noice still. But it looks alot more like it compared to $g[n]$ from a). The poles look to be identical to the two filters poles, just now combined into one filter.

Big code dump:

PYTHON

```python
from typing import final

import matplotlib
matplotlib.use("QtAgg")

from scipy.fft import fft
from scipy.signal import freqz, lfilter, convolve
import matplotlib.pyplot as plt
import numpy as np

L = 500
A_x = A_y = 0.25
f_x = 0.04
f_y = 0.10
n = np.arange(0, L)

d = A_x * np.cos(2*np.pi*f_x*n) + A_y * np.cos(2*np.pi*f_y*n)
e = np.random.normal(size=L)
g = d + e

D_mag = abs(fft(x=d, n=2048)) / 2048
G_mag = abs(fft(x=g, n=2048)) / 2048
freq_steps = np.arange(0, 2048)


fig, (ax1, ax2,) = plt.subplots(2,2, figsize=(15,8))

ax1[0].stem(n, d, label=r"$d[n]$")
ax1[1].plot(freq_steps, D_mag, label=r"$|D(f)|$", c="blue")
ax1[1].set_xlabel(r"$f$")
ax1[1].set_ylabel(r"$|D(f)|$",)
ax1[0].set_ylabel(r"$d[n]$",)
ax1[0].set_xlabel(r"n")



ax2[0].stem(n, g, label=r"$g[n]$", linefmt="r")
ax2[1].plot(freq_steps, G_mag, label=r"$|G(f)|$", c="orange")
ax2[0].set_xlabel(r"$n$")
ax2[0].set_ylabel(r"$g[n]$")
ax2[1].set_xlabel(r"$f$")
ax2[1].set_ylabel(r"$|G(f)|$",)

fig.suptitle("Task 4 a)")
fig.tight_layout(pad=1.0)
fig.legend()



w0_x = 0.04 * 2 * np.pi
w0_y = 0.10 * 2 * np.pi
r = 0.95
```

```
b = [1,0,-1]
a_x = [1, -2*r*np.cos(w0_x), r**2]
a_y = [1, -2*r*np.cos(w0_y), r**2]


w_x, h_x = freqz(b, a_x, 2048)
w_y, h_y = freqz(b, a_y, 2048)

zeros = np.roots(b)
poles_x = np.roots(a_x)
poles_y = np.roots(a_y)

fig3, (ax) = plt.subplots()
theta = np.linspace(-np.pi, np.pi, 1000)
ax.plot(np.sin(theta), np.cos(theta), '--k')
ax.set_aspect(1)
ax.plot(np.real(poles_x), np.imag(poles_x), "Xb", label = r"Poles $H_x(z)$")
ax.plot(np.real(poles_y), np.imag(poles_y), "Xg", label = r"Poles $H_x(z)$")
ax.plot(np.real(zeros), np.imag(zeros), "or", label = "Zeros")

fig40, xa1 = plt.subplots()
#xa1.plot(w_x, 20 * np.log10(abs(h_x)), 'blue', label=r"$H_x$")
#xa1.plot(w_y, 20 * np.log10(abs(h_y)), 'orange', label=r"$H_y$")
xa1.plot(w_x, abs(h_x), 'blue', label=r"$H_x$")
xa1.plot(w_y, abs(h_y), 'orange', label=r"$H_y$")
xa1.set_xlabel(r"$\omega$")
xa1.set_ylabel(r"$|H(e^{j\omega})|$")

tick_positions = [0, np.pi/2, np.pi]
tick_labels = [r'$0$', r'$\frac{\pi}{2}$', r'$\pi$']
xa1.set_xticks(tick_positions)
xa1.set_xticklabels(tick_labels)
fig40.legend()


fig3.suptitle("Task 4 b)")
fig3.legend()
ax.grid()



q_x = lfilter(b, a_x, g,)
q_y = lfilter(b, a_y, g,)

Q_x = abs(fft(x=q_x, n=2048)) / 2048
Q_y = abs(fft(x=q_y, n=2048)) / 2048


fig4, (ax5, ax6,) = plt.subplots(2,2, figsize=(15,8))

ax5[0].stem(n, q_x, label=r"$q_x[n]$")
ax5[1].plot(freq_steps, Q_x, label=r"$|Q_x(f)|$", c="blue")
```

```
ax5[1].set_xlabel(r"$f$")
ax5[1].set_ylabel(r"$|Q_x(f)|$",)
ax5[0].set_ylabel(r"$q_x[n]$",)
ax5[0].set_xlabel(r"n")



ax6[0].stem(n, q_y, label=r"$q_y[n]$", linefmt="r")
ax6[1].plot(freq_steps, Q_y, label=r"$|Q_y(f)|$", c="orange")
ax6[0].set_xlabel(r"$n$")
ax6[0].set_ylabel(r"$q_y[n]$")
ax6[1].set_xlabel(r"$f$")
ax6[1].set_ylabel(r"$|Q_y(f)|$",)

fig4.suptitle("Task 4 c)")
fig4.tight_layout(pad=1.0)
fig4.legend()


b_combined = convolve(b,b)
a_combined = convolve(a_x, a_y)

zeros_combined = np.roots(b_combined)
poles_combined = np.roots(a_combined)


q = lfilter(b_combined, a_combined, g)

Q = abs(fft(x=q, n=2048)) / 2048

w, h_q = freqz(b_combined, a_combined, 2048)


# pole-zero, q[n], |Q(f)|, |H(w)|

final_fig, (axx1, axx2) = plt.subplots(2,2, figsize=(15,8))
axx1[0].stem(n, q, label=r"$q[n]$", )
axx1[0].plot(n, q, linestyle="dashed", c="purple")
axx1[0].set_xlabel("n")
axx1[0].set_ylabel("q[n]")

axx1[1].plot(freq_steps, Q, label=r"$|Q(f)|$", c="green")
axx1[1].set_xlabel("f")
axx1[1].set_ylabel(r"$|Q(f)|$")

axx2[0].plot(w, abs(h_q), label=r"$|H(e^{j\omega})|$", c="orange")
axx2[0].set_xlabel(r"$\omega$")
axx2[0].set_ylabel(r"$|H(e^{j\omega})|$")
axx2[0].set_xticks(tick_positions)
axx2[0].set_xticklabels(tick_labels)

axx2[1].plot(np.sin(theta), np.cos(theta), '--k')
axx2[1].set_aspect(1)
```

```
axx2[1].plot(np.real(poles_combined), np.imag(poles_combined), "Xb", label =
"Poles")
axx2[1].plot(np.real(zeros_combined), np.imag(zeros_combined), "or", label =
"Zeros")
axx2[1].grid()
axx2[1].set_xlabel("Re")
axx2[1].set_ylabel("Im")

final_fig.suptitle("Task 4 d)")
final_fig.legend()
final_fig.tight_layout(pad=1.0)
final_fig.savefig(("taks_4_d.png"))


plt.show()
```